

8(a)(i)

any **two** from:

- 1 higher rate of activity (of rubisco) in GM rice (throughout) ;
- 2 (low CO₂ concentrations) rate of activity for both increases ;
- 3 comparative figures – GM and non-GM at one CO₂ concentration ;

atmospheric CO ₂ concentration / mg dm ⁻³ ±5	rate of activity of rubisco / arbitrary units	
	non-GM rice ±0.05	GM-rice ±0.05
0	0.7	0.80
100	1.8	2.70
200	2.4	3.95
280	2.6	4.15
300	2.6	4.15
400	2.6	4.30
500	2.6	4.45
600	2.6	4.55

3

Question	Answer	Marks
8(a)(i)	<i>plus levels off because:</i> 4 carbon dioxide <u>concentration</u> no longer limiting or light <u>intensity</u> limiting or temperature limiting or rubisco active sites saturated (with carbon dioxide) or less RuBP available ;	
8(a)(ii)	active site ; A allosteric site	1
8(b)(i)	carbon (dioxide) fixation ; A carboxylation	1
8(b)(ii)	reduction / hydrogenation ;	1
8(b)(iii)	6 / 5 ;	1
8(b)(iv)	<i>any two from: list rule</i> 1 hexose / named ; e.g. glucose / fructose 2 disaccharide / named ; e.g. sucrose / maltose 3 starch / cellulose ; 4 amino acids ; 5 glycerol / fatty acids ;	2